



Rules / Productions – OSP-5 Notes

Definition

Rules (or Productions) in Artificial Intelligence are simple **if-then statements** that represent **knowledge** in a structured, logical form. They are widely used in **expert systems**, **decision-making**, and **problem-solving**.

Structure of a Rule

A rule consists of two main parts:

- **IF (Condition / Antecedent):** The situation or pattern to be tested.
- **THEN (Action / Consequent):** The result or action to be taken if the condition is true.

Example:

csharp

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IF the sky is cloudy AND it is cold

THEN it might rain.

Purpose

- To encode **domain knowledge** in a format that can be used by an AI system.
- To support **automated reasoning** and **inference**.



- To allow the system to draw conclusions from available facts.

Types of Rules

1. Deterministic Rules:

- The output is certain if the condition is true.
- Example: IF water reaches 100°C THEN it boils.

2. Probabilistic Rules:

- Include probabilities or uncertainty.
- Example: IF patient has fever THEN 70% chance of infection.

Inference Mechanisms

AI systems use **inference engines** to apply rules and draw conclusions:

Type	Description
Forward Chaining	Starts from known facts and applies rules to reach conclusions.
Backward Chaining	Starts from a goal and works backward to see if facts support it.

Applications

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Area	Use Case
Expert Systems	Diagnosing diseases, legal reasoning
Robotics	Making decisions based on sensor input
Game AI	Deciding moves based on game state
Industrial Control	Monitoring and responding to system conditions

Advantages

- Easy to understand and write.
 - Modular: New rules can be added without major changes.
 - Transparent: Reasoning is traceable and explainable.
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Limitations

- Can become complex with too many rules.
 - May require conflict resolution when multiple rules apply.
 - Not ideal for uncertain or incomplete knowledge unless extended.
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Summary Table

Feature	Description
Format	IF condition THEN action

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Used In	Expert systems, decision-making
Execution Methods	Forward & backward chaining
Benefit	Rule-based logic and inference